

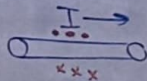
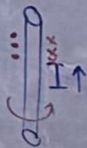
Right Hand Rule

Right Hand Rule:

it is used to determine

- ① current direction
- ② magnetic field direction

B lines form circles around a wire

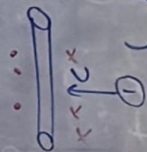


Right Hand Rule:

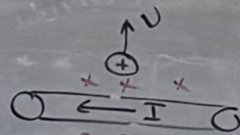
it is used to determine:

- ① velocity direction
- ② magnetic field direction
- ③ magnetic force direction

current moves up

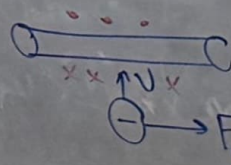
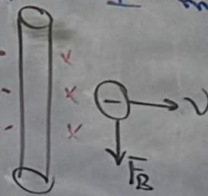


F_B at $+y$



F_B at $-x$

I moves up



I moves right

ملاحظات

F_E vs F_B

Right Hand Rule

① F_E is parallel or antiparallel to E

② F_B is perpendicular to B

③ F_E acts on a charge in E (moving - stationary)

F_B acts on a moving charge in B .

③ F_E does work to displace a charge

F_B does not work to displace a charge

→ F_B can not change the magnitude of v .

→ F_B can change the direction of v

مسألة الكهربية

ex. 2.1 ^{والكترونات}

① 5.3 MeV proton moves upward in $B = 1.2 \text{ mT}$ outward. The magnetic force is.

$\downarrow$ ^ع $\rightarrow$ ^{على}

$$K.E = 5.3 \times 10^6 \times 1.6 \times 10^{-19} = 8.48 \times 10^{-13} \text{ J}$$

$$eV \times 1.6 \times 10^{-19} \rightarrow \text{J}$$

$$v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times 8.48 \times 10^{-13}}{1.67 \times 10^{-27}}} = 3.19 \times 10^7 \text{ m/s}$$

$$v: +y$$

$$B: +z$$

$$F_B: x$$

$$F_B = qvB = 1.6 \times 10^{-19} \times 3.19 \times 10^7 \times 1.2 \times 10^{-3} = 6.125 \times 10^{-15} \text{ N at } +x$$

مسألة الاختبار

الكثافة
ex. 2.1

Right Hand Rule
على

5.3 (e) electron moves upward in $B = 1.2 \text{ mT}$ outward. The magnetic force is.

مسائل ۱۳۰۱

ex. 2.2

(a) A proton moves with $\vec{v} = 2\hat{i} + 3\hat{j}$ in $B = 0.03\hat{i} - 0.15\hat{j}$. The magnetic force is

$$\vec{F}_B = q (\vec{v} \times \vec{B})$$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 0.03 & -0.15 & 0 \end{vmatrix} = \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(-0.3 - 0.09) = -0.39\hat{k} \rightarrow \textcircled{0.39}$$

$$\vec{F}_B = q (\vec{v} \times \vec{B}) = 1.6 \times 10^{-19} \times 0.39 = 0.624 \times 10^{-19} \text{ N at } -z$$

مسائل امتحانی

ex. 2.2

(a) An electron moves with $\vec{v} = 2\hat{i} + 3\hat{j}$ in $B = 0.03\hat{i} - 0.15\hat{j}$. The magnetic force is

$$\vec{F}_B = q (\vec{v} \times \vec{B})$$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 0.03 & -0.15 & 0 \end{vmatrix} = \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(-0.3 - 0.09) = -0.39\hat{k} \rightarrow \textcircled{0.39}$$

$$\vec{F}_B = q (\vec{v} \times \vec{B}) = 1.6 \times 10^{-19} \times 0.39 = 0.624 \times 10^{-19} \text{ N at } z$$

مغناطيسية

a magnetic force on a moving charge:

$$\vec{F} = (q) \vec{v} \times \vec{B} \text{ or } F_B = (q) v B \sin \theta$$

a magnetic force on current carrying conductor:

$$\vec{F} = (I) \vec{L} \times \vec{B} \text{ or } F_B = (I) L B \sin \theta$$

$$q v B = I L B$$

$$(q v = I L)$$

$$e \cdot \frac{m}{s} = \frac{e}{s} \cdot m$$

$$\frac{e}{s} \cdot m = \frac{e}{s} \cdot m$$

As electron

ex. 2.3

0.075m Wire lies along $+x$ Carries 2.4A in $B = 1.6 \hat{k} \text{ (T)}$. The magnetic force is

$$L = 0.075 \text{ (+x)}$$

$$I = 2.4 \text{ A}$$

$$B = 1.6 \text{ (+z)}$$

$$F_B = ILB = 2.4 \times 0.075 \times 1.6 = \boxed{0.288 \text{ N}} \text{ (-y)}$$

or

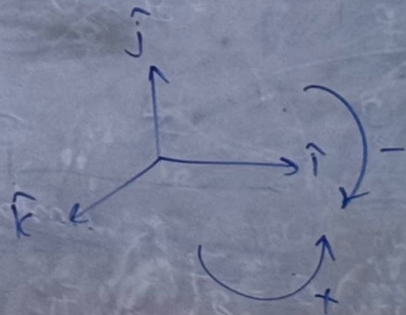
$$\vec{F}_B = I \vec{L} \times \vec{B}$$

$$= 2.4 (0.075 \hat{i} \times 1.6 \hat{k})$$

$$= 2.4 \times 0.075 \times 1.6 (\hat{i} \times \hat{k})$$

$$= 0.288 (-\hat{j})$$

$$= 0.288 \text{ N at -y}$$



مسألة 2.3

ex. 2.3

0.075 (m) Wire lies along $-y$ Carries 2.4 (A) in $B = 1.6\text{ T at } -z$. The magnetic force is

$$L = 0.075 (-y)$$

$$I = 2.4\text{ A}$$

$$B = 1.6 (-z)$$

$$F_B = ILB = 2.4 * 0.075 * 1.6 = \boxed{0.288\text{ N}} (+x)$$

or

$$\vec{F}_B = I \vec{L} \times \vec{B}$$

$$= 2.4 (0.075 \hat{j}) \times 1.6 \hat{k}$$

$$= 2.4 * 0.075 * 1.6 (\hat{j} \times \hat{k})$$

$$= 0.288 (\hat{i})$$

$$= 0.288\text{ N at } +x$$

